

ROANOKE-BLACKSBURG, VA, USA

WMO: 724110

Lat: 37.317N

Lon: 79.974W

Elev: 1175

StdP: 14.08

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13741

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.5	19.8	-3.7	4.7	22.2	0.9	6.0	25.0	27.5	37.3	25.1	34.9	9.1	310	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	92.1	72.6	89.7	71.9	87.5	71.2	75.4	86.3	74.4	84.7	73.3	83.0	8.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.3	124.8	79.3	71.3	120.6	78.4	70.3	116.6	77.8	39.5	86.9	38.5	85.1	37.6	83.3	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	19.4	17.0	9.3	96.0	5.8	3.0	5.1	98.1	1.7	99.9	-1.6	101.6	-5.8	103.7
			7.1	77.5	5.3	1.0	3.3	78.2	0.1	78.8	-2.9	79.3	-6.7	80.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	37.2	40.5	48.0	57.9	65.9	73.5	76.9	75.8	69.2	58.8	48.1	40.7	(6)
	DBStd	16.02	10.33	9.59	9.74	8.47	7.35	5.37	4.58	4.59	6.21	7.80	8.62	8.66	(7)
	HDD50	1347	416	289	157	23	1	0	0	0	15	137	309		(8)
	HDD65	3927	863	687	533	242	82	7	1	0	30	221	509	752	(9)
	CDD50	4197	19	23	94	261	494	706	835	799	576	289	80	21	(10)
	CDD65	1300	0	0	5	30	110	262	371	335	156	30	1	0	(11)
	CDH74	10750	1	3	44	327	905	2112	3219	2752	1142	227	17	1	(12)
	CDH80	3740	0	0	3	71	254	723	1270	1023	350	46	0	0	(13)
(14) Wind	WSAvg	6.5	7.9	7.9	8.2	7.8	6.4	5.8	5.4	4.9	5.0	5.6	6.4	6.8	(14)
(15) Precipitation	PrecAvg	41.6	2.9	3.0	3.5	3.3	4.1	3.9	4.1	3.8	3.7	3.3	3.1	3.0	(15)
	PrecMax	62.4	8.0	8.0	7.9	11.3	10.1	10.3	12.8	9.5	11.7	9.9	12.4	8.2	(16)
	PrecMin	25.0	0.3	0.6	0.4	0.5	1.0	0.6	0.4	0.7	0.2	0.0	0.2	0.2	(17)
	PrecStd	8.1	1.8	1.7	1.7	2.0	2.1	2.4	2.1	1.9	2.8	2.2	2.0	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	71.3	79.1	86.3	89.4	93.4	96.0	95.3	91.1	84.8	76.7	69.4	(19)
		MCWB	57.3	58.1	60.3	63.1	68.9	72.9	73.4	72.9	70.5	67.6	61.2	57.2	(20)
	2%	DB	63.0	65.5	73.9	81.8	85.8	89.7	92.3	91.4	87.7	79.9	70.5	63.4	(21)
		MCWB	55.5	53.1	57.3	61.6	68.2	71.9	73.2	72.6	69.6	65.3	57.5	55.0	(22)
	5%	DB	58.2	60.5	69.0	77.7	82.9	87.0	89.6	88.6	84.1	75.7	66.4	59.0	(23)
		MCWB	50.3	50.1	54.5	60.1	67.1	71.0	72.8	72.0	68.5	63.3	56.4	51.8	(24)
	10%	DB	53.0	56.1	64.5	73.5	79.4	84.4	87.0	85.8	80.7	72.0	62.5	54.7	(25)
		MCWB	45.7	46.8	52.3	58.6	65.5	69.8	72.0	71.1	67.5	61.6	53.6	47.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.0	60.8	62.6	67.8	72.7	75.5	77.1	76.7	74.1	72.3	64.9	61.9	(27)
		MCDB	65.3	68.6	74.9	77.7	83.6	87.0	89.1	88.4	83.9	79.2	70.3	67.4	(28)
	2%	WB	56.6	55.8	59.8	64.8	70.6	74.0	75.6	75.2	72.5	68.9	61.6	57.1	(29)
		MCDB	62.2	62.1	70.0	76.1	81.4	85.1	86.7	85.8	81.3	75.2	67.8	61.1	(30)
	5%	WB	51.6	51.9	57.3	62.6	69.0	72.9	74.6	74.0	71.2	66.1	58.7	53.1	(31)
		MCDB	56.3	58.1	66.0	73.0	79.0	83.4	85.2	83.8	79.2	73.2	64.3	57.6	(32)
	10%	WB	46.1	47.4	54.4	60.4	67.4	71.6	73.5	73.0	69.8	63.3	55.1	48.2	(33)
		MCDB	52.2	55.1	62.5	70.3	76.3	81.3	83.3	82.0	77.2	70.0	61.1	53.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	17.6	19.4	21.1	19.9	18.7	18.2	18.2	18.6	19.5	19.1	15.8	(35)
		MCDBR	22.1	25.3	26.2	26.9	23.9	21.9	21.4	21.7	23.4	24.1	25.2	22.1	(36)
		MCWBR	15.5	16.8	14.6	12.5	10.4	8.6	7.2	7.3	9.0	11.9	15.2	15.4	(37)
	5% WB	MCDBR	20.0	22.9	21.9	22.4	20.7	19.3	18.6	18.5	18.2	18.6	20.4	19.5	(38)
		MCWBR	16.2	17.0	14.0	12.1	9.9	8.3	7.3	7.1	8.0	10.6	15.1	16.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.304	0.321	0.350	0.396	0.436	0.477	0.507	0.506	0.400	0.360	0.322	0.305	(40)
	taud		2.494	2.432	2.401	2.293	2.192	2.116	2.057	2.032	2.329	2.433	2.503	2.520	(41)
	Ebn at Noon		282	290	290	281	270	258	249	246	269	272	272	273	(42)
	Edh at Noon		25	30	35	41	46	50	53	53	37	30	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		718	948	1280	1609	1778	1937	1825	1661	1387	1106	823	625	(44)
	RadStd		58	114	82	126	154	136	116	83	137	133	72	55	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (42 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+275	+164
		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+137

Nomenclature: See separate page